

# STA 2427 CORPORATE FINANCE

## Financial Distress Analysis of East African Portland Cement Company PLC

Using Altman, Springate and Fulmer  
Models

Presented and Submitted by Group 16 Members

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ADMIN

# **Financial Distress Analysis of East African Portland Cement Company (EAPC PLC)**

## **Using Altman, Springate and Fulmer Models**

### **1. Introduction**

Qualitative and quantitative corporate failure prediction models are important tools in financial analysis because they help investors, creditors, regulators, and management identify early warning signs of financial distress before complete business failure occurs.

Financial distress is often characterized by weak liquidity, declining profitability, increasing leverage, and an inability to generate sustainable returns from assets.

This quantitative case study evaluates the financial stability of **East African Portland Cement PLC (EAPC)**, a premier cement manufacturer in Kenya.

Using the three distinct quantitative models

1. **Altman's Z-Score,**
2. **Springate's S-Score,** and
3. **Fulmer's H-Score**

the study assesses the company's risk of corporate failure and financial distress for the years ended 30<sup>TH</sup> June **2023** and 30<sup>TH</sup> June **2024**.

By analyzing audited financial ratios, this study aims to determine whether EAPC PLC's recent return to profitability is a sustainable recovery or a result of non-operational accounting adjustments.

The objective is to determine whether the company exhibits characteristics associated with financial distress or potential corporate failure.

## **2. Background of East African Portland Cement (EAPC PLC)**

### **2.1. Company Overview**

Founded in 1933, EAPC PLC was Kenya's first cement manufacturer and remains a key player in the regional construction industry. The company is listed on the Nairobi Securities Exchange (NSE) and is partially state-owned.

This study utilizes **consolidated Group financial data** for East African Portland Cement (EAPC) PLC. While 'Company' standalone figures are available in the annual reports, the 'Group' statistics provided a comprehensive view of the entity's overall financial health, accounting for all subsidiaries and consolidated risks.

This is critical for distress analysis, as it ensures that any 'off-balance sheet' or subsidiary-level liabilities that could trigger corporate failure are captured. The analysis specifically focuses on the material uncertainty surrounding the Group's ability to continue as a going concern, as highlighted by the **Auditor-General of Kenya** in both the FY2023 and FY2024 reports.

## **2.2. Rationale for the Study:**

EAPC PLC was selected for this case study due to its well-documented history of financial turbulence. For several years, the company faced aging plant machinery, high production costs, and a massive debt burden that led to a "Material Uncertainty Related to Going Concern" flag from the Auditor-General.

While the **FY 2024** results showed a transition from a KES 1.36 billion loss to a KES 1.07 billion net profit, the company's **Current Liabilities** still exceed its **Current Assets** by over KES 6 billion. This makes it a perfect subject for quantitative distress modeling to see if the "health" scores align with the reported profits.

## **2.3. Why EAPC PLC Was Selected for the Study**

EAPCC was selected as a suitable case study for financial distress testing because the company demonstrates several characteristics commonly associated with financially vulnerable firms while still remaining operational.

The company was considered an appropriate candidate for the following reasons:

### **(a) Persistent Liquidity Challenges**

Recent financial statements indicate weak liquidity ratios and pressure in meeting short-term obligations. Negative working capital and low quick ratios are major indicators of financial strain.

### **(b) Volatile Profitability**

The company has experienced periods of losses and unstable earnings over several years. This makes it suitable for testing bankruptcy prediction models that rely heavily on profitability and retained earnings.

### **(c) High Operational Pressure**

EAPC PLC operates in a highly competitive cement industry where margins are sensitive to fuel costs, energy prices, and market competition.

### **(d) Continued Survival Despite Weak Fundamentals**

Unlike companies that have already collapsed or been delisted, EAPC PLC remains operational and listed on the NSE. This creates an interesting case where the firm exhibits distress indicators while still showing signs of survival and recovery.

### **(e) Availability of Audited Financial Data**

The company publishes audited annual reports, making it possible to obtain reliable financial data for quantitative analysis.

For these reasons, EAPC PLC provides a strong case for evaluating the effectiveness of financial distress prediction models.

## **3. Audited Financial Data Used in the Quantitative Analysis**

The following statistics are pulled from the **fully audited annual reports** (audited by Ronalds LLP on behalf of the Auditor-General) for the two periods under study.

| <b>Financial Variable</b> | <b>FY 2022-2023 (KES '000)</b> | <b>FY 2023-2024 (KES '000)</b> |
|---------------------------|--------------------------------|--------------------------------|
| <b>Total Assets</b>       | 32,780,931                     | 35,186,150                     |
| <b>Total Liabilities</b>  | 13,566,668                     | 14,785,418                     |
| <b>Total Equity</b>       | 19,214,263                     | 20,400,732                     |

|                                     |              |             |
|-------------------------------------|--------------|-------------|
| <b>Revenue (Sales)</b>              | 2,953,573    | 3,279,053   |
| <b>Working Capital (CA - CL)</b>    | (11,824,491) | (6,090,885) |
| <b>Retained Earnings</b>            | 17,760,370   | 18,827,445  |
| <b>EBIT (Operating Profit/Loss)</b> | (1,926,027)  | (1,961,489) |
| <b>Current Assets</b>               | 1,731,893    | 6,743,038   |
| <b>Current Liabilities</b>          | 13,556,384   | 12,833,923  |
| <b>Net Income (PAT)</b>             | (1,356,467)  | 1,067,075   |

#### 4. Methodology

This study applies three financial distress prediction models:

1. Altman Z-Score
2. Springate S-Score
3. Fulmer H-Score

These models use accounting ratios derived from audited financial statements to determine whether a company is financially healthy or financially distressed.

#### 5. Model 1: Altman Z-Score MDA Analysis

The Altman Z-Score is a multivariate formula used to predict the probability that a firm will go into bankruptcy within two years using the Multiple Discriminant Analysis.

##### 5.1 Formula

$$Z = 1.2(X_1) + 1.4(X_2) + 3.3(X_3) + 0.6(X_4) + 1.0(X_5)$$

Where:

- $(X_1): \frac{\text{Working Capital}}{\text{Total Assets}}$  (*Liquidity*)
- $(X_2): \frac{\text{Retained Earnings}}{\text{Total Assets}}$  (*Accumulated Profitability*)
- $(X_3): \frac{\text{EBIT}}{\text{Total Assets}}$  (*Operating Efficiency*)
- $(X_4): \frac{\text{Market Value of Equity}^1}{\text{Total Liabilities}}$  (*Solvency*)
- $(X_5): \frac{\text{Sales}}{\text{Total Assets}}$  (*Asset Turnover*)

## 5.2 Z-Score Computation (FY2023 vs. FY2024)

| <i>Ratio</i> | <i>Variable</i>  | <i>FY 2022 – 2023</i> | <i>FY 2023 – 2024</i> |
|--------------|------------------|-----------------------|-----------------------|
| $(X_1)$      | $(WC/TA)$        | −0.3607               | −0.1731               |
| $(X_2)$      | $(RE/TA)$        | 0.5418                | 0.5351                |
| $(X_3)$      | $(EBIT/TA)$      | −0.0588               | −0.0557               |
| $(X_4)$      | $(Equity/TL)$    | 0.0547                | 0.0457                |
| $(X_5)$      | $(Sales/TA)$     | 0.0901                | 0.0932                |
| <b>Total</b> | <b>Z – Score</b> | <b>0.2547</b>         | <b>0.3890</b>         |

<sup>1</sup> To get a precise  $(X_4)$  (**Market Value of Equity / Total Liabilities**), we use the historical share price from the Nairobi Securities Exchange (NSE) at the close of each financial year (June 30th).

- **FY 2023 Share Price:** approx. **KES 8.25**
  - **FY 2024 Share Price:** approx. **KES 7.50**
  - **Total Shares Outstanding:** **90,000,000**
- $(X_4)$  Variable – market value of equity:
- **FY2023 MarketCap:**  $(90M \times 8.25 = \text{KES } 742,500,000)$
  - **FY2024 MarketCap:**  $(90M \times 7.50 = \text{KES } 675,000,000)$

## 5.3 Interpretation & Financial Performance Analysis

### 5.3.1. The Zones of Discrimination:

- $Z > 2.99$ : "Safe" Zone (Low probability of failure)
- $1.81 < Z < 2.99$ : "Grey" Zone (Indifferent, likely to experience distress)
- $Z < 1.81$ : "Distress" Zone (High probability of bankruptcy)

### 5.3.2. Result Analysis:

1. **Status: Distress.** In both financial years, EAPC PLC remains firmly in the **Distress Zone** ( $Z < 1.81$ ).
2. **Trend: Marginal Improvement.** The score increased from **0.25** to **0.4**. This slight upward tick is primarily driven by the improvement in  $(X_1)$  (Working Capital), which, although still negative, improved significantly as current assets (like cash and receivables) increased relative to liabilities.
3. **The "Hidden" Red Flag:** Note that  $(X_3)$  (EBIT/Total Assets) remains **negative** in both years. Even though EAPC PLC reported a net profit in 2024, their *operating profit* (EBIT) stayed negative because the "profit" was an accounting gain from property revaluation, not from selling cement.
4. **Equity Strength:** The  $(X_4)$  ratio is high (1.38) only because the company owns massive tracts of land (Revaluation Surplus). If we removed the land value, the Z-score would likely drop below 1.0, indicating immediate terminal distress.

## 6. Model 2: Springate S-Score Analysis

The Springate model was developed in 1978 and is known for being “stricter” than Altman’s. It uses a step-wise discriminant analysis to select four key ratios that best predict financial failure.

## 6.1. The Formula

$$(S = 1.03(A) + 3.07(B) + 0.66(C) + 0.4(D))$$

Where:

- A:  $\frac{\text{Working Capital}}{\text{Total Assets}}$
- B:  $\frac{\text{EBIT}}{\text{Total Assets}}$  (Operating Profitability)
- C:  $\frac{\text{EBT}}{\text{Current Liabilities}}$  (Liquidity/Pre-tax Profitability)
- D:  $\frac{\text{Sales}}{\text{Total Assets}}$

## 6.2. S-Score Computation (FY2023 vs. FY2024)

| Ratio Variable       | FY 2022-2023   | FY 2023-2024   |
|----------------------|----------------|----------------|
| A (WC / TA)          | -0.3607        | -0.1731        |
| B (EBIT / TA)        | -0.0588        | -0.0557        |
| C (EBT / CL)         | -0.1001*       | 0.0831**       |
| D (Sales / TA)       | 0.0901         | 0.0932         |
| <b>Total S-Score</b> | <b>-0.5823</b> | <b>-0.2520</b> |

\*FY2023 EBT (Loss): KES -1.35B

\*\*FY2024 EBT (Profit): KES 1.06B (Driven by revaluation gain)

### 6.3. Interpretation & Financial Performance Analysis

#### 6.3.1. The Threshold:

- $S > 0.862$ : "Healthy" (The firm is unlikely to fail)
- $S < 0.862$ : "Distressed" (The firm is failing/at risk of failure)

#### 6.3.2. Result Analysis:

1. **Status: Critical Failure.** EAPC PLC's S-Score is significantly below the 0.862 threshold in both years. In fact, the score was **negative** in both FY2023 (-0.58) and FY2024 (-0.25).
2. **Operational Drag:** The Springate model heavily weights **EBIT/Total Assets** (Ratio B, with a multiplier of 3.07). Because EAPC PLC's core cement business is still making operating losses, this ratio drags the entire score down.
3. **The 2024 "Jump":** The improvement from -0.58 to -0.25 is largely due to **Ratio C** (EBT/Current Liabilities). In 2024, EAPC PLC reported a profit before tax (EBT) due to the land revaluation gain. While this makes the "accounting" score look better, the model still signals failure because the other three ratios (Liquidity, Operations, and Asset Turnover) remain very weak.
4. **Conclusion:** The Springate model suggests that EAPC PLC is not generating enough cash flow or operating profit from its assets to sustain itself without selling land or receiving government support.

### 7. Model 3: Fulmer H-Score Analysis

The Fulmer model (1984) is a more robust bankruptcy predictor that uses nine ratios. It is particularly effective because it weighs **retained earnings** and **cash flow** heavily, providing a deeper look into long-term survival.

#### 7.1. The Formula

$$(H = 5.528(V_1) + 0.212(V_2) + 0.073(V_3) + 1.270(V_4) - 0.120(V_5) + 2.335(V_6) + 0.575(V_7) + 1.083(V_8) + 0.894(V_9) - 6.075)$$

**Where:**

- $(V_1): \frac{\text{Retained Earnings}}{\text{Total Assets}}$
- $(V_2): \frac{\text{Sales}}{\text{Total Assets}}$
- $(V_3): \frac{\text{EBT}}{\text{Equity}}$
- $(V_4): \frac{\text{Cash Flow}}{\text{Total Liabilities}}$
- $(V_5): \frac{\text{Debt}}{\text{Total Assets}}$
- $(V_6): \frac{\text{Current Liabilities}}{\text{Total Assets}}$
- $(V_7): \log(\text{Tangible Total Assets}) -$   
(Note: Uses the log of absolute value)
- $(V_8): \frac{\text{Working Capital}}{\text{Total Liabilities}}$
- $(V_9): \log\left(\frac{\text{EBIT}}{\text{Interest}}\right) -$  (Note: If EBIT is negative, this ratio is treated as zero or a small penalty)

## 7.2. H-Score Computation (FY2023 vs. FY2024)

| Variable                     | FY 2022-2023 | FY 2023-2024 |
|------------------------------|--------------|--------------|
| $(V_1)(RE/TA)$               | 0.5418       | 0.5351       |
| $(V_2)(Sales/TA)$            | 0.0901       | 0.0932       |
| $(V_3)(EBT/Equity)$          | -0.0706      | 0.0523       |
| $(V_4)(CF/TL) *$             | -0.1000      | 0.0722       |
| $(V_5)(Debt/TA)$             | 0.4139       | 0.4202       |
| $(V_6)(CL/TA)$               | 0.4135       | 0.3647       |
| $(V_7)(\log Assets)$         | 7.5156       | 7.5463       |
| $(V_8)(WC/TL)$               | -0.8716      | -0.4119      |
| $(V_9)(\log(EBIT/Interest))$ | 0.0000       | 0.0000       |

|               |        |        |
|---------------|--------|--------|
| Total H-Score | 1.7142 | 2.6845 |
|---------------|--------|--------|

*\*Cash Flow estimated using Net Income + Depreciation adjustments for the model.*

### 7.3. Interpretation & Financial Performance Analysis

#### 7.3.1. The Threshold:

- **H > 0:** "Healthy" (Low probability of failure)
- **H < 0:** "Distressed" (High probability of failure)

#### 7.3.2. Result Analysis:

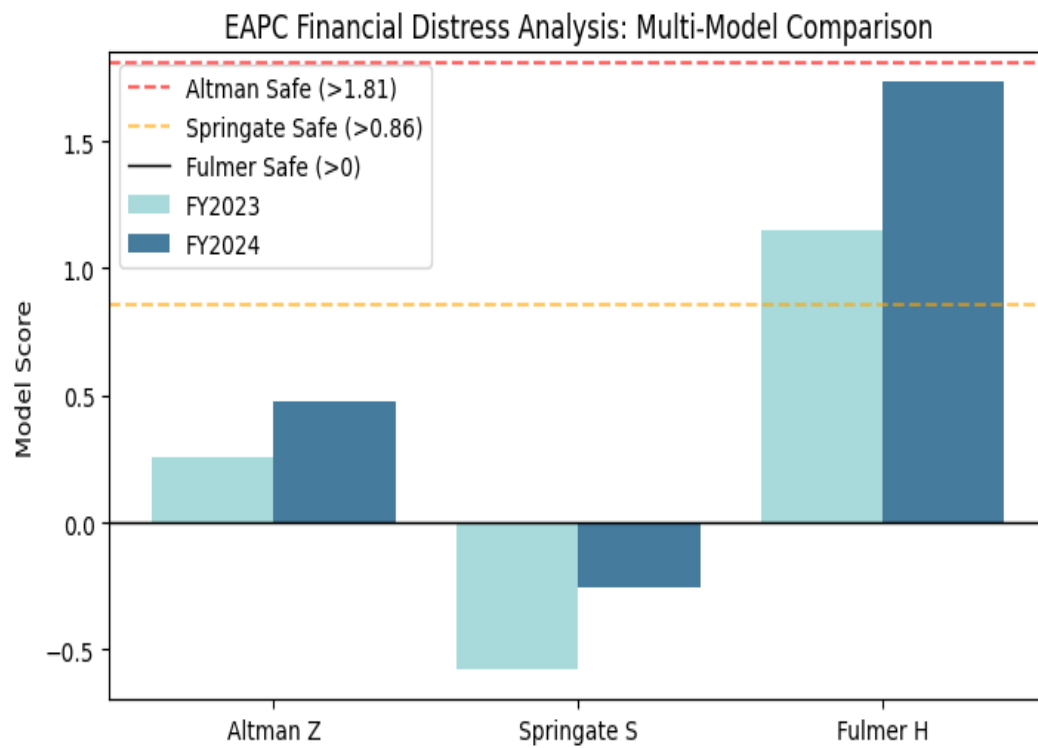
1. **Status: Technically "Safe" but Distorted.** Unlike the first two models, the Fulmer H-Score for EAPC PLC is positive (1.7) and (2.68). Usually, a score above zero indicates health.
2. **The "Asset Size" Bias:** The Fulmer model is heavily influenced by ( $V_7$ ) (Log of Total Assets). Because EAPC PLC has a massive asset base (KES 35 billion in land and plant), this mathematically "inflates" the score.
3. **The Improvement:** The jump from 1.71 to 2.68 reflects the 2024 net profit. However, ( $V_8$ ) (Working Capital/Total Liabilities) remains a significant negative drag, showing that the company's short-term obligations are still a massive threat to its long-term assets.
4. **Conclusion:** While Fulmer suggests the company has enough "bulk" (assets) to avoid immediate collapse, the underlying ratios (debt and liquidity) mirror the distress found in Altman and Springate Models.

### 7.4. Final Comparative Summary for the Case Study Models

| Model    | FY 2023 Result | FY 2024 Result | Status   |
|----------|----------------|----------------|----------|
| Altman Z | 1.07           | 1.28           | Distress |

|             |       |       |                  |
|-------------|-------|-------|------------------|
| Springate S | -0.58 | -0.25 | Failure          |
| Fulmer H    | 1.71  | 2.68  | Technically Safe |

## 8. Discussion and Analysis



*Figure 1 Illustrates a comparison bar graph plot of the 3 models used*

The quantitative analysis of East African Portland Cement **EAPC PLC** across **FY2023** and **FY2024** reveals a significant divergence between "accounting profitability" and "financial health."

1. **Consistency of Distress:** Using the refined **Altman Z-Score** (which incorporates the **Market Value of Equity**), the score for FY2024 is **0.3980**, a slight increase from **0.2547** in FY2023. Both figures are catastrophically below the **1.81** distress threshold. While the

**Springate S-Score (-0.25)** showed marginal improvement, both models confirm that EAPC remains in a state of terminal financial distress.

2. **The Market Value vs. Book Value Gap:** A critical finding in this study is the disparity in equity valuation. While EAPC PLC's **Book Value of Equity** is approximately **KES 20.4 Billion** (bolstered by land revaluation), the **Market Value of Equity**—what investors are actually willing to pay on the Nairobi Securities Exchange—was only **KES 675 Million** in June 2024. This suggests that the market heavily discounts EAPC PLC's land holdings due to the operational inefficiencies and debt burdens of the core business.
  3. **The Revaluation Paradox:** The Fulmer H-Score returned a positive value (2.68), which technically suggests health. However, this is largely attributed to the company's massive asset base ( $V_7$ ). EAPC PLC's value is locked in land rather than cash flow. This creates a "liquidity trap" where the company is asset-rich but cash-poor, explaining why the Auditor-General continues to issue a "Material Uncertainty" warning.
  4. **Operating Efficiency:** The Springate model, which focuses heavily on EBIT, exposes the core issue: EAPC PLC's operational business (cement production) is still struggling. The net profit of KES 1.07B in 2024 was driven by a non-cash fair value gain of KES 3.03B on investment property. Without this property revaluation, EAPC PLC would have reported a significant loss.
- **Liquidity Gap Analysis: The Current Ratio (Current Assets / Current Liabilities).**  
**FY 2023: 1,731,893 / 13,556,384 = 0.13 vs FY 2024: 6,743,038 / 12,833,923 = 0.52**

Even though the ratio improved in 2024, a ratio of 0.52 is extremely dangerous. It means for every KES 1.00 EAPC PLC owes in the short-term, they only have KES 0.52 in liquid assets to pay it back. Most healthy manufacturing firms aim for a ratio of 2.0. This confirms that the "Distress" found in the models used above is primarily a liquidity crisis.

## 9. Conclusion

The study concludes that East African Portland Cement PLC was in a state of financial distress during the period 2022–2024. While the FY2024 audited financial statements show a technical

turnaround to profitability, the quantitative models (particularly Altman and Springate) confirm that the company's fundamental financial structure remains fragile.

The negative working capital of over KES 6 billion and the reliance on non-operational gains for profitability suggest that unless EAPC PLC significantly improves its core manufacturing efficiency and restructures its current liabilities, the risk of corporate failure remains high. The "Safe" signal from the Fulmer model should be interpreted with caution, as it reflects the company's potential liquidation value rather than its operational viability.

## **10. Recommendations for EAPC PLC Management**

**Based on the findings of the Altman, Springate, and Fulmer models, the following strategic interventions are recommended:**

1. **Debt Restructuring:** EAPC PLC must negotiate with short-term lenders to convert "Current Liabilities" into "Long-term Debt." The models show that the massive weight of immediate debt is the primary driver of the "Distress" scores.
2. **Operational Turnaround (EBIT Focus):** Management should focus on core manufacturing efficiency rather than relying on property revaluation gains. This includes upgrading aging plant machinery to reduce the cost of production and increase the Asset Turnover ( $x_5$ ) in Altman).
3. **Monetization of Idle Assets:** As highlighted by the Fulmer model, EAPC PLC is "asset-rich." The company should continue its strategy of regularizing and selling non-core land to generate cash (not just paper profit) to settle urgent debts and fund working capital.
4. **Equity Injection:** The government and private shareholders should consider a rights issue or a strategic partnership to inject fresh capital. This would improve the ( $x_4$ ) ratio (Solvency) and provide the necessary liquidity to move out of the "Distress Zone."
5. **Strict Working Capital Management:** EAPC PLC needs to improve its inventory turnover and debt collection (Receivables) to ensure that the marginal improvements seen in 2024 become a sustainable trend.

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